

V71 Test Sequence Settings — Engineering Record

Instrument: Vitrek V71 AC/DC Hipot Safety Tester **Station:** Juniper Automated Test Station (hi - pot_UART controller) **Recorded:** 17 August 2026 **Revision:** 1 **Source:** Transcribed from the V71 front panel (EDIT SEQUENCE and CONFIG MENU screens)

Purpose and status of this document

This is the written record of what the V71 is configured to do. The instrument's communication protocol is **write-only for test sequences** — there is no command that reads a stored sequence back out, and the unit has no file export. If the V71 is reset, replaced, or its memory is corrupted, this document and the matching `import_panel_sequences.py` in the controller repository are the only way the sequences come back.

The test values were set by the responsible electrical engineer. They are recorded here exactly as found. Nothing in this document authorises a change to them. Any change is the engineer's decision, is made on the instrument, and is then re-recorded here at a new revision.

External equipment is connected to the DUT and sits in the ground / continuity measurement path. Readings therefore reflect the fixture as a whole rather than a bare conductor.

Summary

#	Sequence name	Steps	Purpose
1	CONT TEST	1	Continuity check only
2	HIGH POT	1	AC dielectric withstand only
3	CONT HIGHPOT	2	Continuity, then dielectric withstand

Sequence 3 is sequences 1 and 2 run back to back. Because **ON FAIL** is set to *Stop Running Sequence*, a continuity failure in step 1 prevents the 1500 V step from running.

Sequence 1 — CONT TEST

One step. Continuity measurement, no high voltage applied.

Panel field	Setting
TYPE	CONT
DWELL	5 sec
LIMITS	5 kΩ max (no minimum)
ZERO	0 Ω

Controller command: ADD,CONT,5.0,,5000.0

A reading at or below 5 kΩ passes. The empty field between the two commas is the minimum limit, left as NONE.

Sequence 2 — HIGH POT

One step. AC dielectric withstand at 1500 V.

Panel field	Setting
TYPE	ACW
LEVEL	1500 V
DUT	Isolated
RAMP	1 sec
DWELL	1 sec
LIMITS	Breakdown Only

Controller command: ADD,ACW,1500.0,1.0,1.0,,

"Breakdown Only" means both leakage limits are set to NONE. Per the V7X manual the instrument still detects breakdown in this state, in accordance with most standards; what it does not do is apply a numeric pass/fail window to the measured leakage current. The two trailing empty fields in the command are those two limits.

RAMP runs before DWELL and is not included in it, so total energised time for this step is approximately 2 seconds.

Sequence 3 — CONT HIGHPOT

Two steps, run in order. Settings are identical to sequences 1 and 2.

Step	Panel field	Setting
1	TYPE	CONT
1	DWELL	5 sec
1	LIMITS	5 k Ω max
1	ZERO	0 Ω
2	TYPE	ACW
2	LEVEL	1500 V
2	DUT	Isolated
2	RAMP	1 sec
2	DWELL	1 sec
2	LIMITS	Breakdown Only

Controller commands:

```
ADD,CONT,5.0,,5000.0
ADD,ACW,1500.0,1.0,1.0,,
```

Global configuration (CONFIG MENU)

These settings apply to **every** sequence and are stored separately from them. They are part of the acceptance criteria even though they do not appear in any sequence definition. The same named sequence run on a differently-configured unit can produce a different pass/fail result.

Setting	Value	Effect
IFACE	USB	Selects the USB (HID-to-UART) port for computer control
VICL	0 Switch Units	No 964 switch matrix attached
DIO	Inputs Off	Digital I/O inputs, including INTERLOCK, are disabled
START	Requires Stop before Start	Front panel START must be preceded by STOP
BEEP	On	Audible indication enabled
LOCK	Disabled	No password required to change sequences or configuration
FREQ	60 Hz	Test frequency for ACW steps
ARC	20 mA	Arcing above 20 mA fails the step
RAMP DOWN	Fast Discharge	Voltage is discharged quickly rather than ramped down
ON FAIL	Stop Running Sequence	A failed step halts the remaining steps

ARC deserves particular attention. It is an active pass/fail criterion applied to every ACW step, but it lives in the CONFIG MENU rather than in the sequence. A `RESET CONFIG` on the instrument would set it to 0 (disabled) and the HIGH POT sequence would quietly stop failing on arcing, with no change to the sequence itself and nothing on screen to indicate the difference.

The controller webapp can capture these settings to a timestamped JSON file (HiPot page → Instrument Config → **Back Up Settings**). Doing so after any configuration change, and keeping the file with the project, is the practical protection against this.

What a backup does not cover

The following cannot be read from the instrument over the interface, by the controller or by any other software, and exist only where they are written down:

- **The test sequences themselves.** Write-only protocol, as described above.
- **IFACE.** No remote command — deliberate, since changing it remotely could sever the connection issuing the command.
- **CONT ZERO and GB ZERO lead-resistance offsets** (UTILITY MENU). No remote command; these must be re-measured by hand after a reset.
- **Per-step ZERO offsets** inside CONT steps. Not settable over the interface at all. The CONT step above uses 0 Ω , so nothing is currently lost to this — but a future sequence that used a non-zero offset could not be fully reproduced by the controller.
- **The CONFIG MENU lock password.**

Reproducing these sequences

On the controller station:

```
python import_panel_sequences.py
```

This writes all three sequences into the station database, from which they appear in the operator dropdown on the HiPot page. The commands generated are verified against the V7X manual's configuration field tables by `test_add_commands.py`.

Automatic first-run seeding. As of the controller build that added `seed_default_sequences()`, the app also seeds these three sequences by itself the first time it starts on a fresh install — so a re-imaged station comes up with the operator dropdown already populated instead of empty (the results database lives in `ProgramData` and does not survive a re-image). The seed is **one-time** (guarded by an `app_settings` marker) and only ever *creates* names that are entirely absent, so it never overwrites or resurrects a sequence an admin has since edited or deleted. Running `import_panel_sequences.py` by hand is still the way to reload or, with `--replace`, overwrite them after a change to this record.

To load a sequence into the instrument itself, run it once from the controller — the controller programs sequence #0 over the interface at run time. Sequence #0 is volatile and is lost at power-off; the numbered stores 1–3 on the front panel are non-volatile and hold the copies recorded in this document.

Revision history

Rev	Date	Change	By
1	2026-08-17	Initial record, transcribed from the instrument front panel	—

Verification

Recorded settings checked against the instrument front panel:

Name: **Date:**

Signature:

Settings authorised by (electrical engineer):

Name: **Date:**

Signature: